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(54) **SYSTEM AND METHOD FOR DISPENSING
POWDER FOR AMMUNITION**

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(57) **ABSTRACT**

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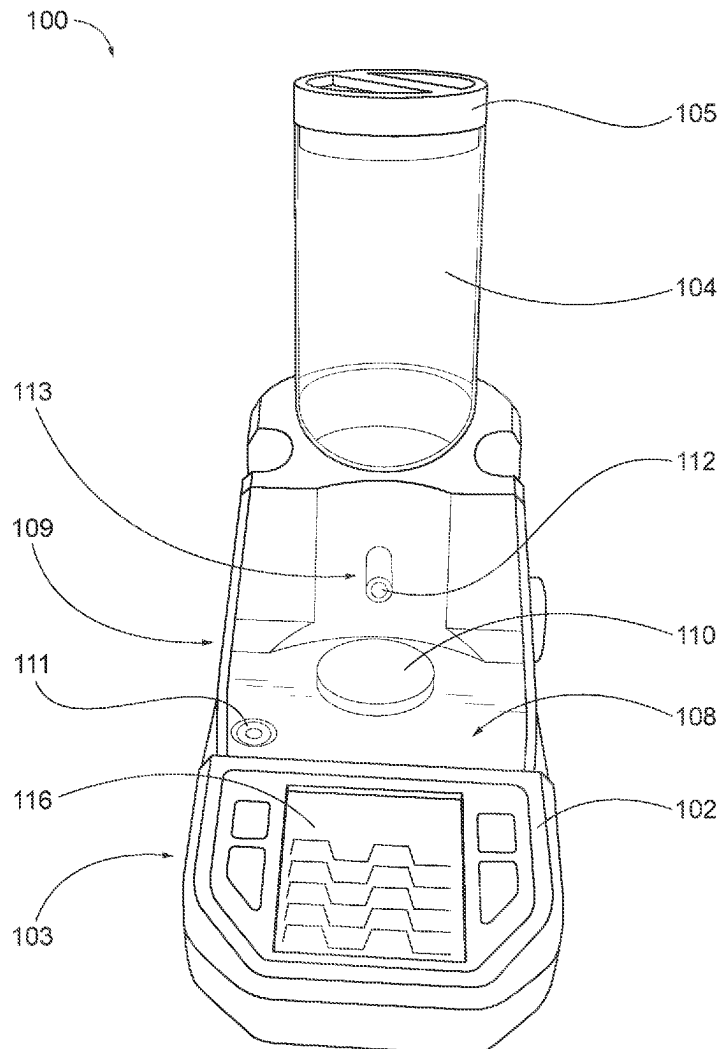
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(63) Continuation of application No. 17/864,839, filed on
Jul. 14, 2022, now Pat. No. 12,326,325.

(60) Provisional application No. 63/221,928, filed on Jul.
14, 2021.

A system, device and method for accurately and efficiently dispensing powder for loading ammunition. The dispenser or loader is configured to autonomously determine the powder type and the dispense rate of powder is determined for selected rotational speeds based on powder type. The most efficient dispensing regime is determined using a number of fixed speeds of barrel rotation wherein switch points among the speeds of barrel rotation, stopping points and trickle rates to approach a target weight without overshooting.



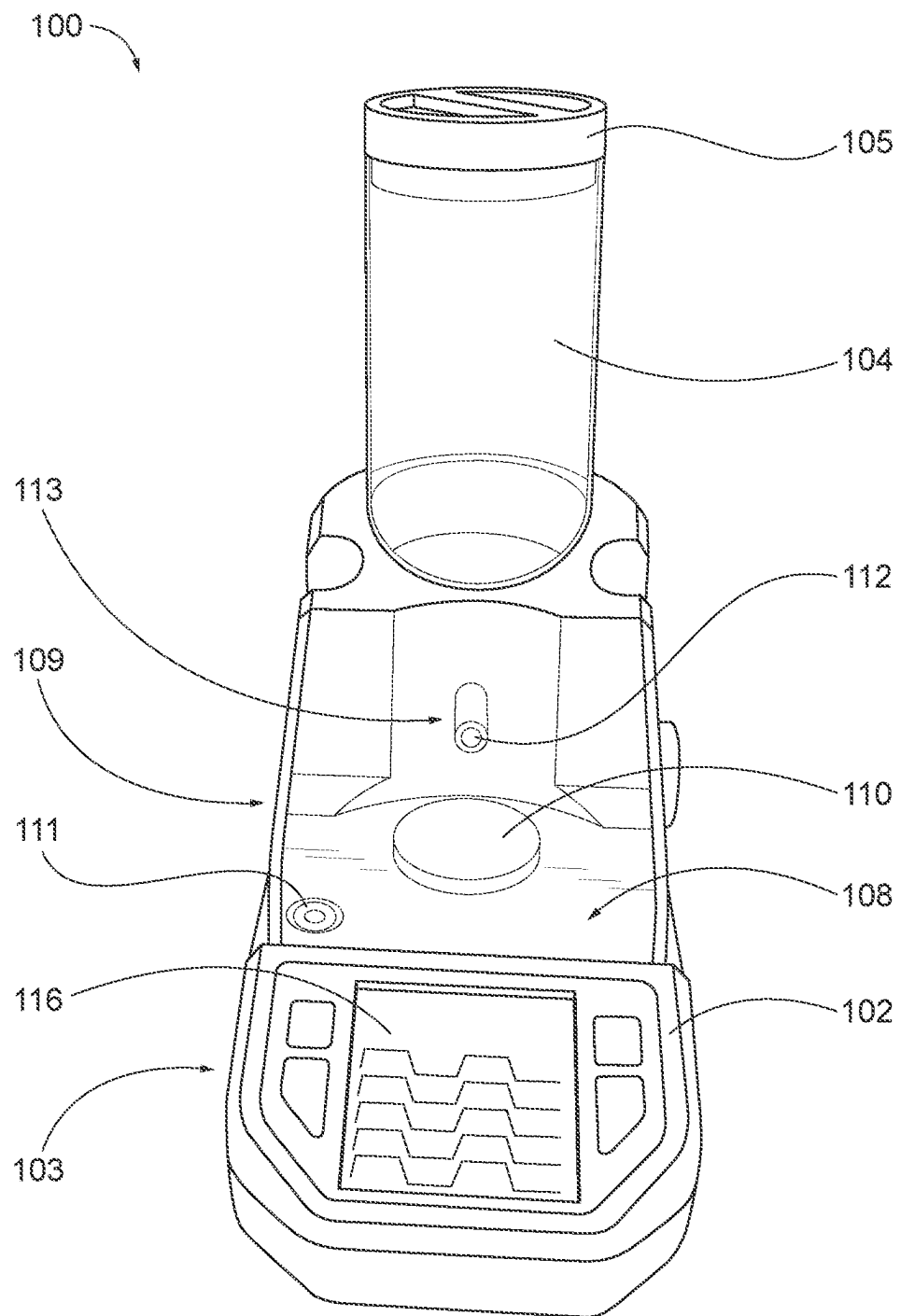


FIG. 1

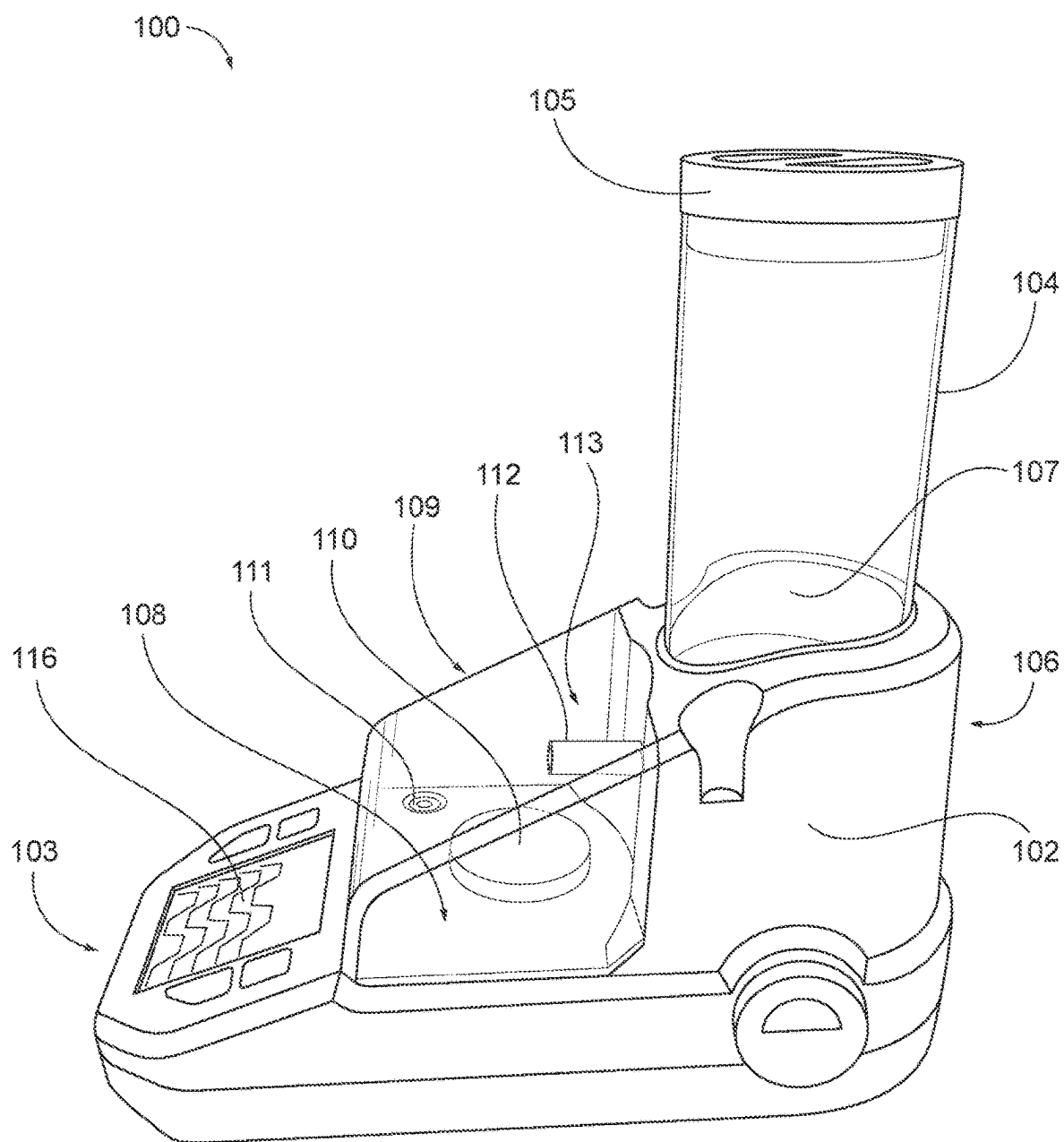


FIG. 2

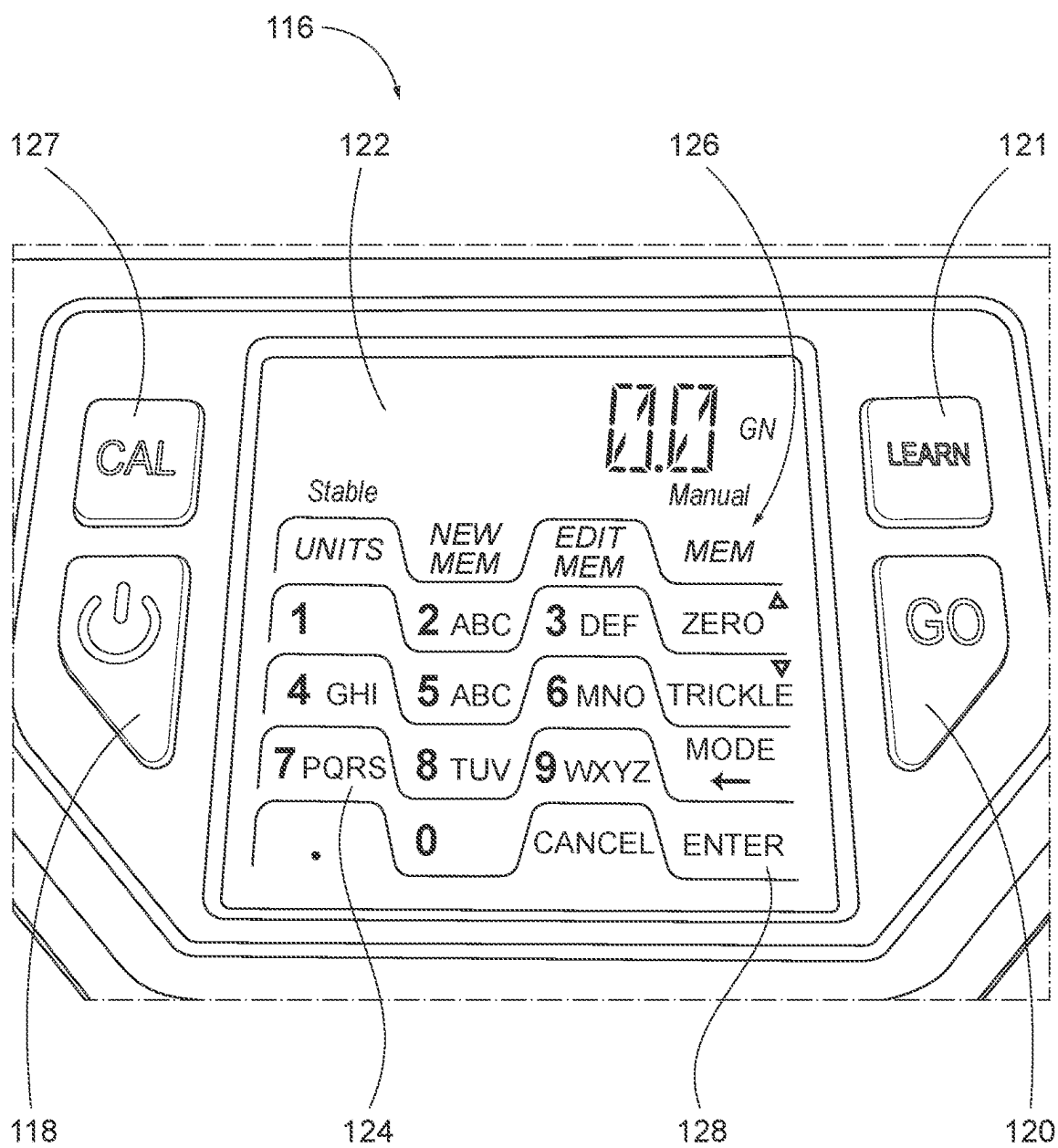


FIG. 3

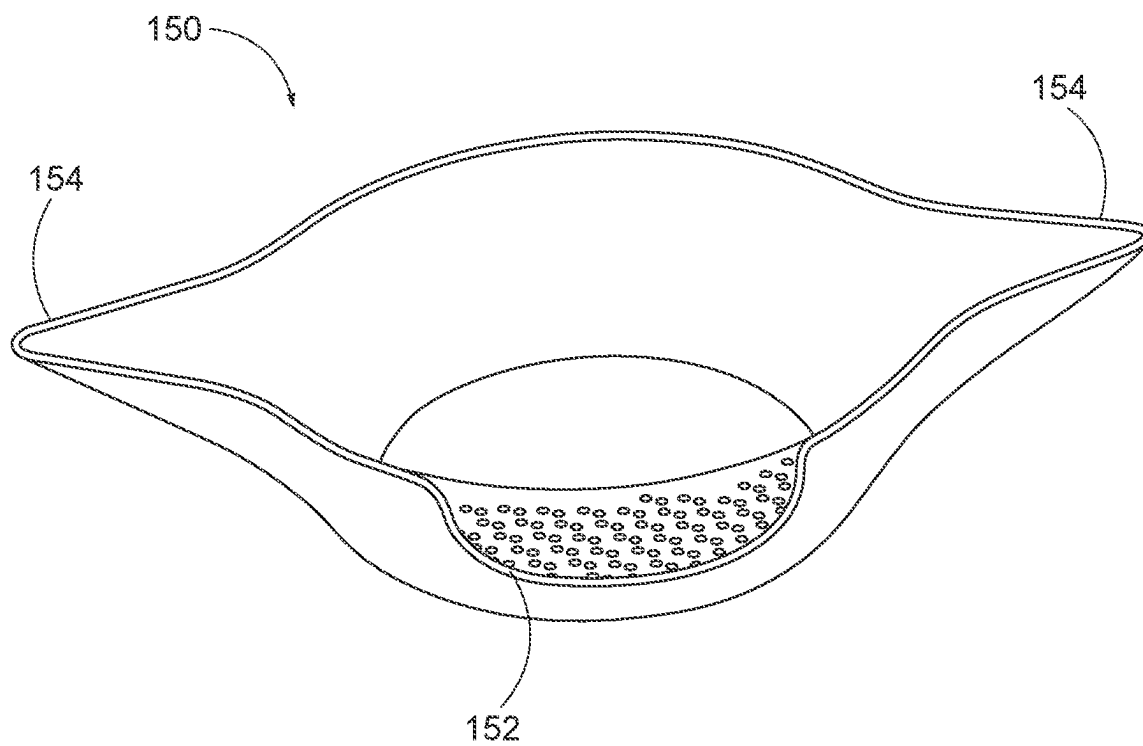


FIG. 4

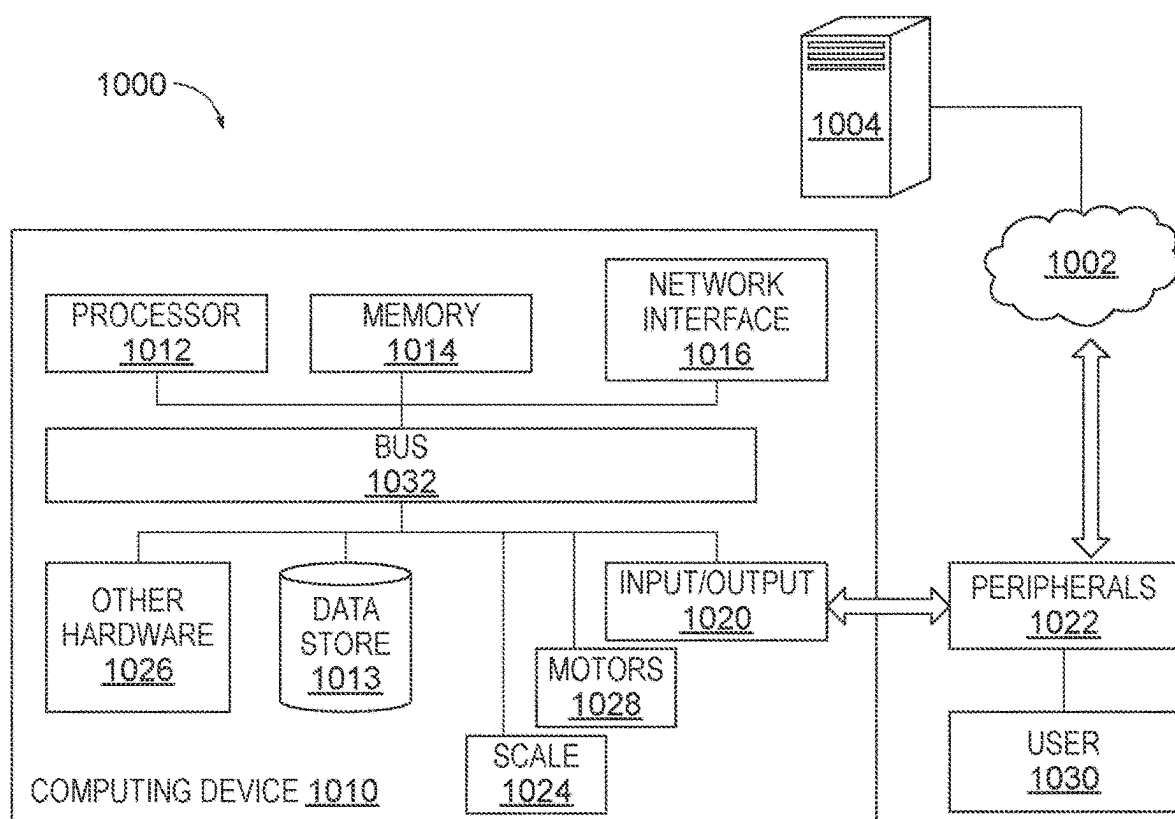


FIG. 5

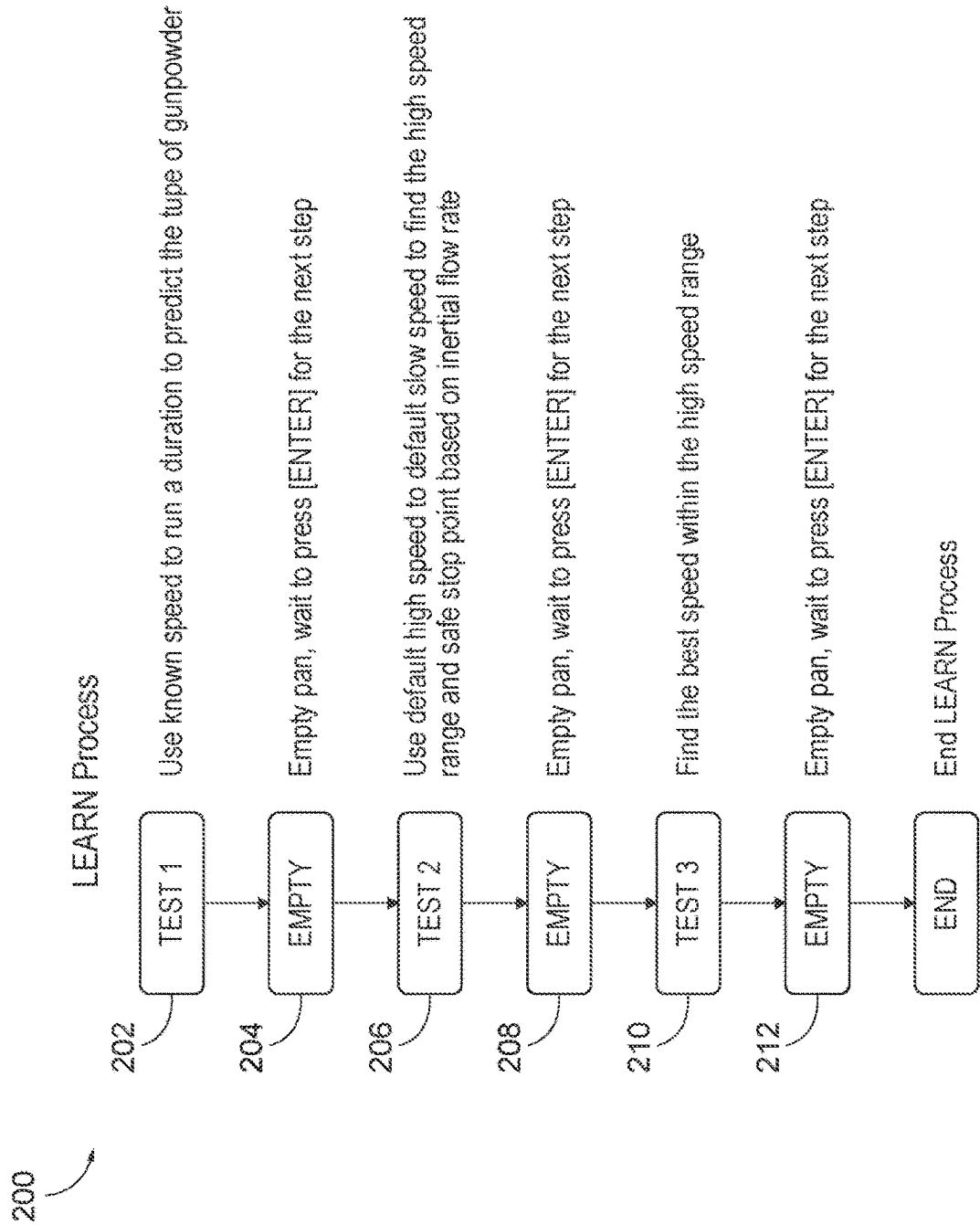


FIG. 6

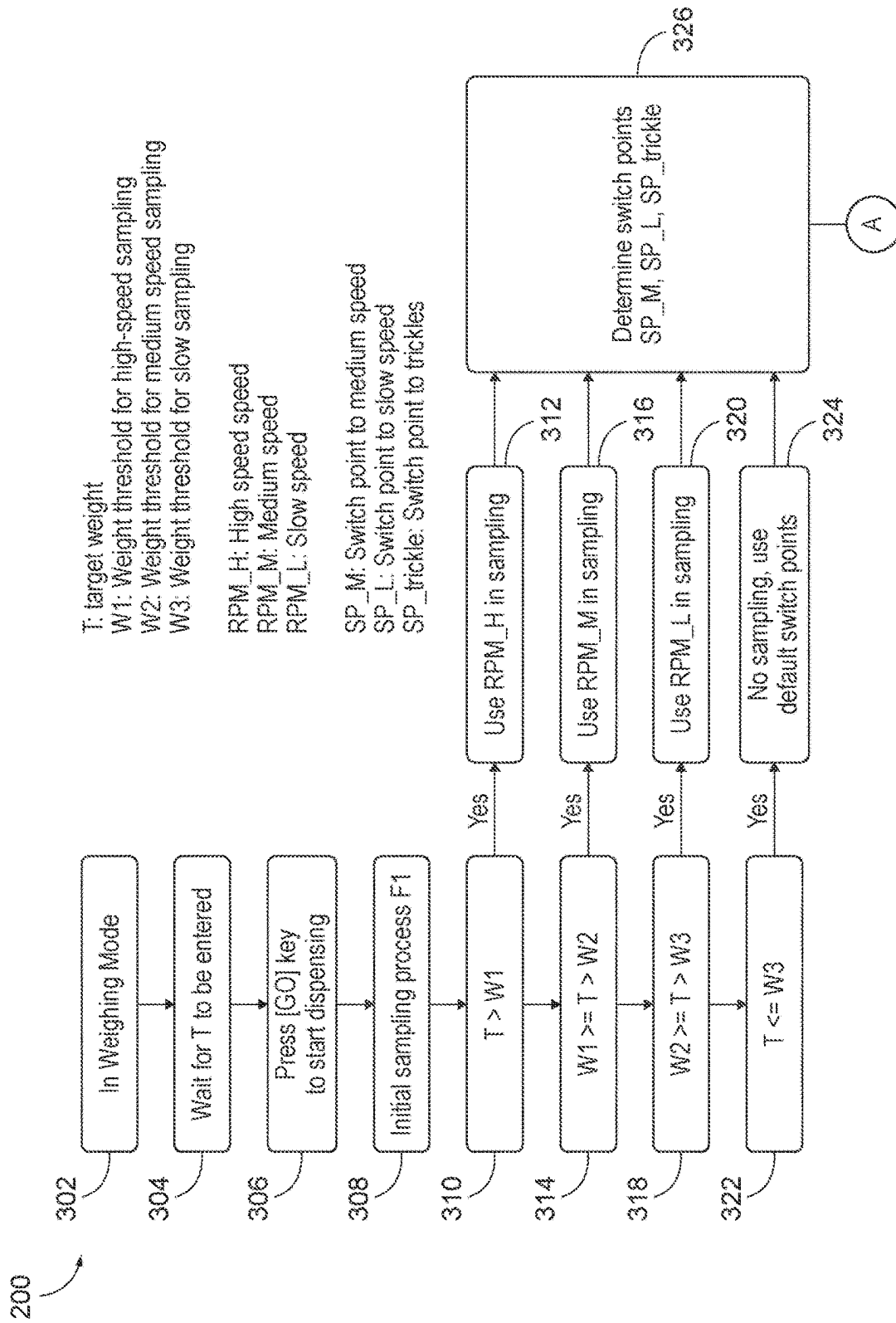


FIG. 7

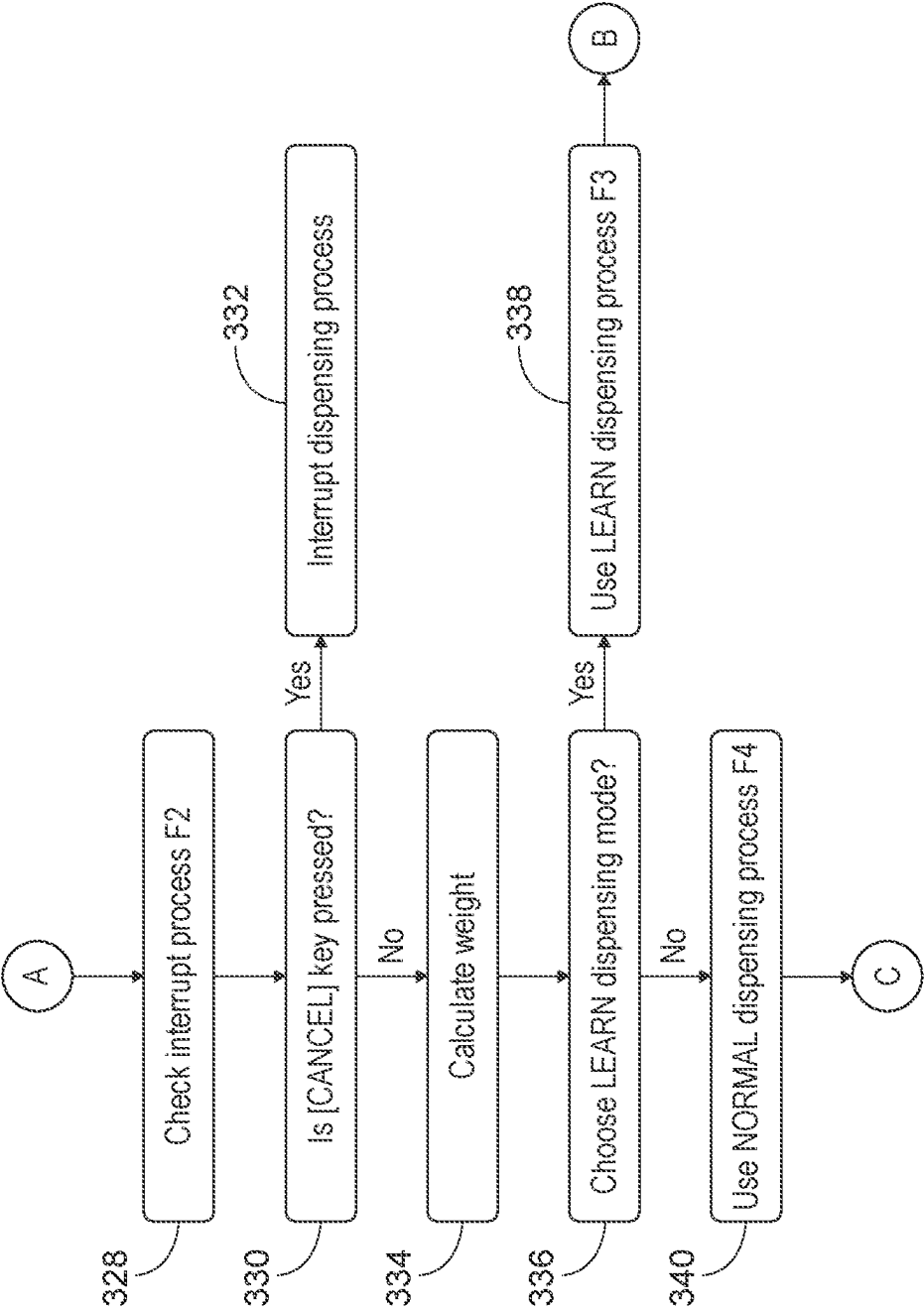


FIG. 8

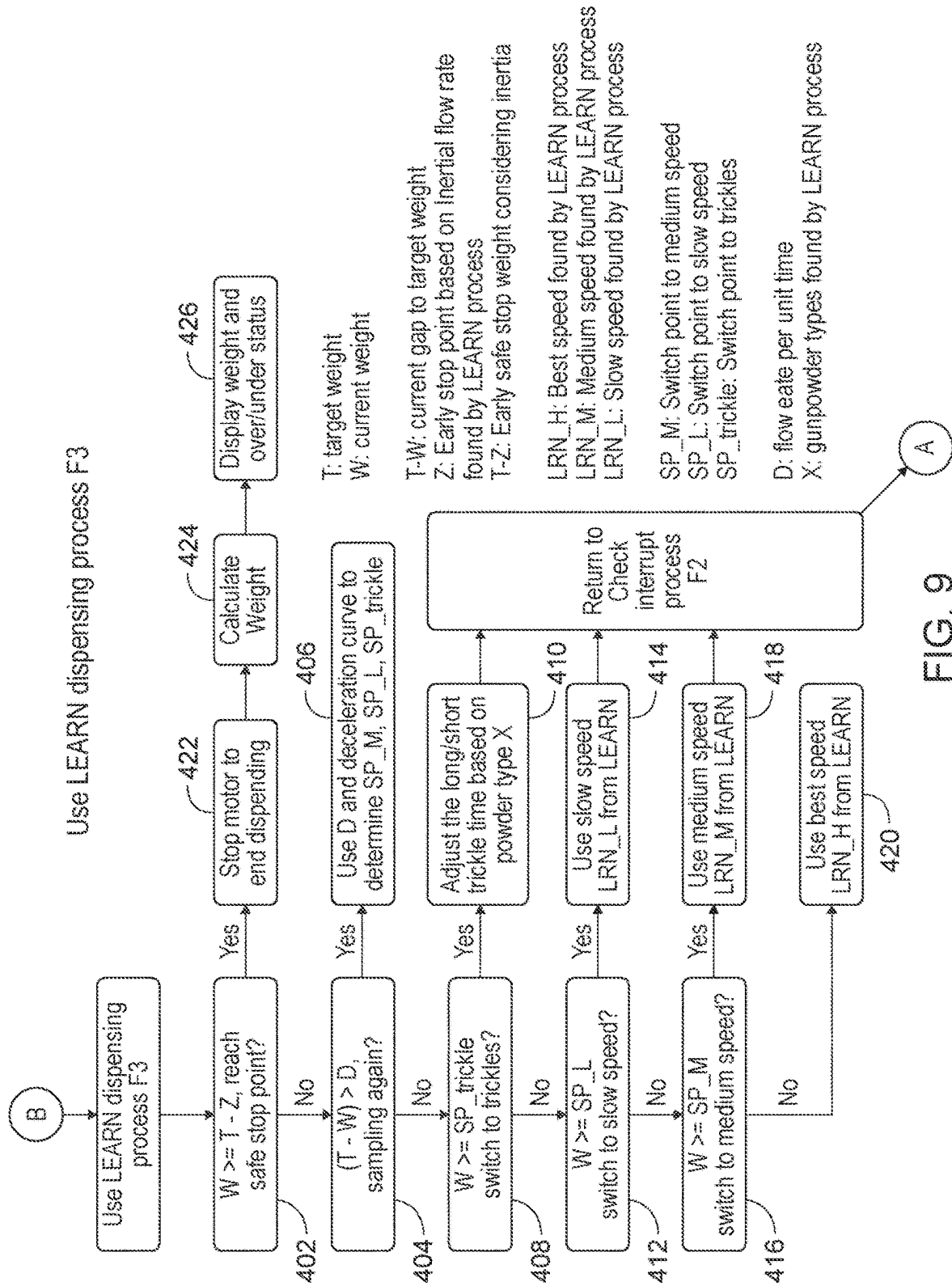
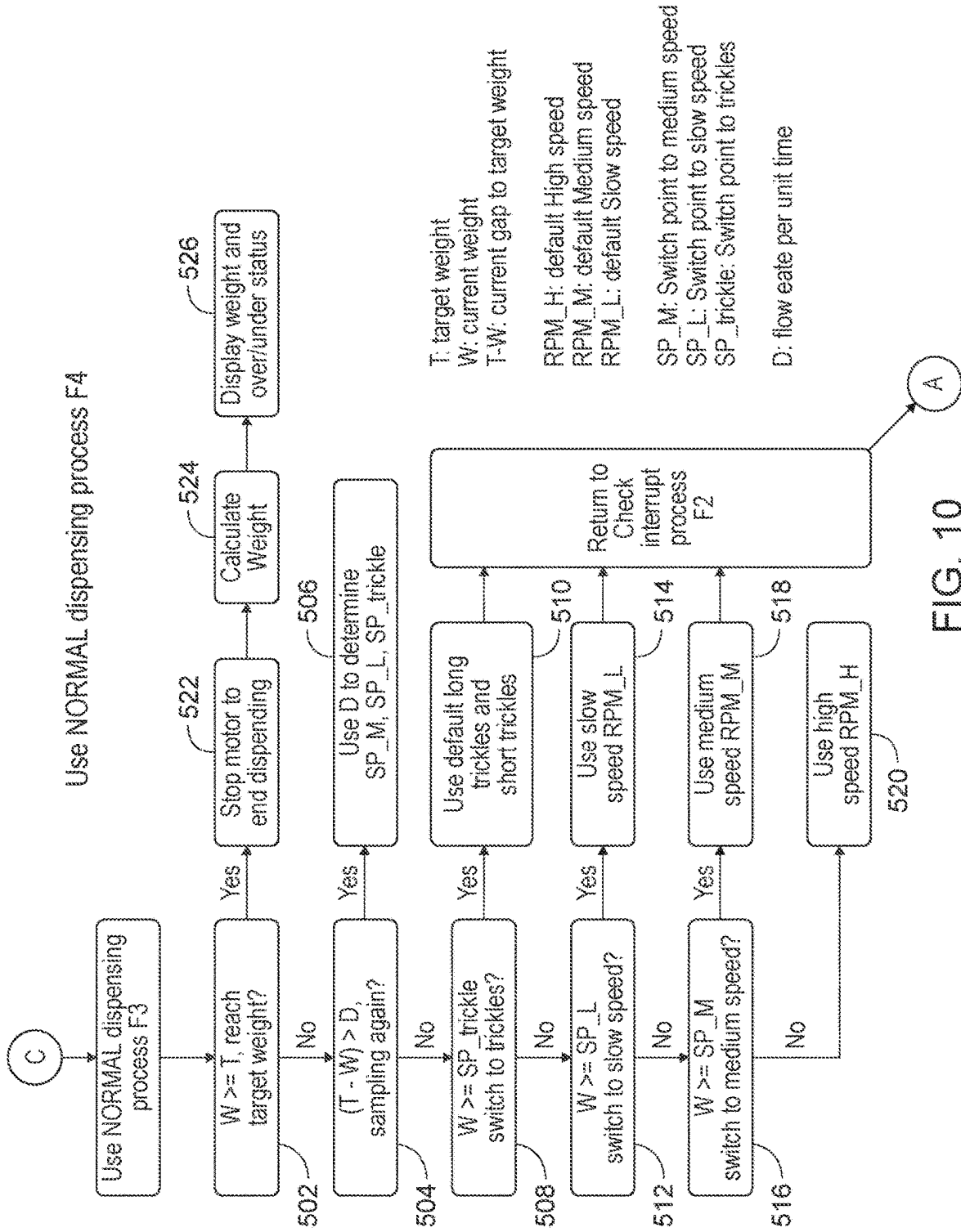


FIG. 9



SYSTEM AND METHOD FOR DISPENSING POWDER FOR AMMUNITION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of application Ser. No. 17/864,839, filed Jul. 14, 2022 that claims the benefit of priority to U.S. patent application Ser. No. 63/221,928 filed Jul. 14, 2021, the contents of which are incorporated herein by reference in their entirety.

FIELD

[0002] The present disclosure relates to a powder dispenser for ammunition loading, and more particularly to a device for a dispensing system with increased accuracy and speed.

BACKGROUND

[0003] Experienced hunters may calibrate their firearm or sighting device for a particular trajectory pattern specific to a given powder weight. Mass-produced and packaged ammunition, however, can have inconsistent and varying powder charge weights for each bullet, even within the same production run and package. As a result, the varying weights typical to commercially produced ammunition can result in inconsistent and inaccurate shooting. These variances may be small and therefore undetectable even by an enthusiast in the field without additional equipment and analysis. Powder dispensers allow outdoor enthusiasts, such as hunters and marksmen, to weigh, fill and load their own ammunition. Each individual powder charge is typically weighed and measured to load the ammunition to a desired weight.

[0004] Highly accurate scales may be used to measure an exact powder weight for a cartridge. The fine granularity of the powder, however, still makes measuring and dispensing the exact weight within a tolerable range difficult. Traditional powder dispensers may use scales, volumes, and other measuring devices to dispense powder quickly, at the cost of accuracy. Other devices may be tuned to slowly dispense a powder for a more accurate volume, however, at the cost of expediency. Previous solutions requires the use of separate and independent devices, which required an enthusiast user to operate the two dispensers independently in series to load a single cartridge. Such ad-hoc systems, however, are time consuming, costly, and cumbersome.

SUMMARY

[0005] According to one aspect, and described herein, a powder dispenser offers a highly-accurate, and highly efficient system and method for dispensing a consistent amount of powder. A processor or controller in communication with a user interface may determine speeds at which a dispenser barrel rotates in order to dispense a volume and weight of powder from a hopper to a receptacle. A scale monitors and determines the total weight of powder dispensed into the receptacle resting on the scale. The entire assembly may be self-contained in a single integrated body.

[0006] The powder dispensing device may include a housing and a hopper sized and shaped to engage the housing to form a reservoir. A first barrel and an optional second barrel may be in fluid communication with the reservoir. At least one motor may be coupled to the first barrel and the optional second barrel and is configured to rotate the first barrel and

the optional second barrel. A scale is electrically coupled to a scale plate. The scale plate may be disposed on the housing and below the first barrel and optional second barrel. A user interface may be in electrical communication with the at least one motor and the scale. The user interface may be configured to receive a user input.

[0007] An aspect of the disclosure relates to a method executable by a processor on the dispenser to perform a method for dispensing powder. The processor may be configured to operate the at least one motor according to a user input and receive a measurement from the scale. The processor may drive the barrel(s) at one or more rotational speeds according to the measurement from the scale.

[0008] The method comprises a processor controlling at least one motor to rotate a barrel or tube at a first rotational speed for a period of time to dispense powder from a hopper to a receptacle on a scale; receiving data from a sensor on the scale, the data comprising an amount of powder dispensed to the receptacle; and determining a category for the powder based on an amount of powder dispensed during the period of time.

[0009] Embodiments of the method include the following, alone or in any combination.

[0010] The method may comprise the processor determining a safe stopping point and a trickle rate regime comprising a long trickle rate and a slow trickle rate based on the category.

[0011] The method may comprise the processor controlling the at least one motor to rotate the barrel or tube at a plurality of rotational speeds to dispense powder from the hopper to the receptacle on a scale, wherein each of the plurality of speeds is run for a period of time; receiving data comprising an amount of powder dispensed to the receptacle for each of the plurality of rotational speeds; determining the dispense rate of powder based on the amount of powder dispensed and the run time for each of the plurality of rotational speeds; and determining a most effective high speed for dispensing powder.

[0012] The method may comprise the processor receiving a target weight from a user via a user interface; controlling the at least one motor to rotate the barrel or tube at one or more speeds to dispense powder based on a difference between the target weight and first, second and third weight parameters, wherein the second weight parameter is smaller than the first weight parameter and the third weight parameter is smaller than the second weight parameter, until the dispensed weight reaches the safe stopping point; and controlling the at least one motor to rotate the barrel or tube at a long trickle rate and a slow trickle rate until the target weight is reached.

[0013] When the target weight is greater than the first weight parameter, the processor may control the at least one motor to rotate the barrel or tube at a most effective high speed for dispensing powder, the most effective high speed determined from the dispense rate of powder based on the amount of powder dispensed and the run time for each of a plurality of rotational speeds; and determines switch points to a default medium speed and a default low speed based on second and third weight parameters that are each smaller than the first weight parameter.

[0014] When the target weight is between the first weight parameter and the second weight parameter, the processor may control the at least one motor to rotate the barrel or tube

at a default medium speed; and determines a switch point to a default low speed based on the second and third weight parameters.

[0015] When the target weight is between the second weight parameter and the third weight parameter, the processor may control the at least one motor to rotate the barrel or tube at a default slow speed.

[0016] Another aspect provides a non-transitory computer readable medium, comprising a plurality of instructions stored thereon, the plurality of instructions configurable to be executed by a processor such that upon execution the processor is configurable to control at least one motor to rotate a barrel or tube at a first rotational speed for a period of time to dispense powder from a hopper to a receptacle on a scale; receive data from a sensor on the scale, the data comprising an amount of powder dispensed to the receptacle; and determine a category for the powder based on an amount of powder dispensed during the period of time.

[0017] Embodiments of the non-transitory computer readable medium include the following, alone or in any combination.

[0018] The processor may determine a safe stopping point and a trickle rate regime comprising a long trickle rate and a slow trickle rate based on the category.

[0019] The processor may control the at least one motor to rotate the barrel or tube at a plurality of rotational speeds to dispense powder from the hopper to the receptacle on a scale, wherein each of the plurality of speeds is run for a period of time; receive data comprising an amount of powder dispensed to the receptacle for each of the plurality of rotational speeds; determine the dispense rate of powder based on the amount of powder dispensed and the run time for each of the plurality of rotational speeds; and determine a most effective high speed for dispensing powder.

[0020] The processor may receive a target weight from a user via a user interface; control the at least one motor to rotate the barrel or tube at one or more speeds to dispense powder based on a difference between the target weight and first, second and third weight parameters, wherein the second weight parameter is smaller than the first weight parameter and the third weight parameter is smaller than the second weight parameter, until the dispensed weight reaches the safe stopping point; and control the at least one motor to rotate the barrel or tube at a long trickle rate and a slow trickle rate until the target weight is reached.

[0021] When the target weight is greater than the first weight parameter, the processor may control the at least one motor to rotate the barrel or tube at a most effective high speed for dispensing powder, the most effective high speed determined from the dispense rate of powder based on the amount of powder dispensed and the run time for each of a plurality of rotational speeds; and determine switch points to a default medium speed and a default low speed based on second and third weight parameters that are each smaller than the first weight parameter.

[0022] When the target weight is between the first weight parameter and the second weight parameter, the processor may control the at least one motor to rotate the barrel or tube at a default medium speed; and determine a switch point to a default low speed based on the second and third weight parameters.

[0023] When the target weight is between the second weight parameter and the third weight parameter, the pro-

cessor may control the at least one motor to rotate the barrel or tube at a default slow speed.

[0024] Another aspect provides a powder dispensing device comprising a housing, a hopper sized and shaped to engage the housing to form a reservoir; a barrel in fluid communication with the reservoir, at least one motor coupled to the barrel, the at least one motor configured to rotate barrel, a scale electrically coupled to a scale plate, the scale plate disposed on the housing and below the barrel; a user interface in electrical communication with the at least one motor and the scale, the user interface configured to receive a user-input; and a processor configured to execute computer readable instructions to control at least one motor to rotate a barrel or tube at a first rotational speed for a period of time to dispense powder from a hopper to a receptacle on a scale; receive data from a sensor on the scale, the data comprising an amount of powder dispensed to the receptacle; determine a category for the powder based on the amount of powder dispensed during the period of time.

[0025] Embodiments of the device include the following, alone or in any combination.

[0026] The device comprises instructions to the processor to determine a safe stopping point and a trickle rate regime comprising a long trickle rate and a slow trickle rate based on the category.

[0027] The device comprises instructions to the processor to control the at least one motor to rotate the barrel or tube at a plurality of rotational speeds to dispense powder from the hopper to the receptacle on a scale, wherein each of the plurality of speeds is run for a period of time; receive data comprising an amount of powder dispensed to the receptacle for each of the plurality of rotational speeds; determine the dispense rate of powder based on the amount of powder dispensed and the run time for each of the plurality of rotational speeds; and determine a most effective high speed for dispensing powder.

[0028] The device comprises instructions to the processor to receive a target weight from a user via a user interface; control the at least one motor to rotate the barrel or tube at one or more speeds to dispense powder based on a difference between the target weight and first, second and third weight parameters, wherein the second weight parameter is smaller than the first weight parameter and the third weight parameter is smaller than the second weight parameter, until the dispensed weight reaches the safe stopping point; and control the at least one motor to rotate the barrel or tube at a long trickle rate, a slow trickle rate or both until the target weight is reached.

[0029] When the target weight is greater than the first weight parameter, the processor controls the at least one motor to rotate the barrel or tube at a most effective high speed for dispensing powder, the most effective high speed determined from the dispense rate of powder based on the amount of powder dispensed and the run time for each of a plurality of rotational speeds; and determines switch points to a default medium speed and a default low speed based on second and third weight parameters that are each smaller than the first weight parameter.

[0030] When the target weight is between the first weight parameter and the second weight parameter, the processor controls the at least one motor to rotate the barrel or tube at a default medium speed; and determines a switch point to a default low speed based on the second and third weight parameters.

[0031] When the target weight is between the second weight parameter and the third weight parameter, the processor controls the at least one motor to rotate the barrel or tube at a default slow speed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] The features, nature, and advantages of the present disclosure will become more apparent from the detailed description set forth below when taken in conjunction with the drawings in which like reference characters identify correspondingly throughout.

[0033] FIG. 1 depicts a front view of a powder dispenser device according to an exemplary embodiment of one aspect of the present disclosure.

[0034] FIG. 2 depicts a side view of a powder dispenser device according to one an exemplary embodiment of aspect of the present disclosure.

[0035] FIG. 3 depicts an expanded view of a user-interface of a powder dispenser device according to an exemplary embodiment of one aspect of the present disclosure.

[0036] FIG. 4 depicts a perspective view of a powder tray according to an exemplary embodiment of one aspect of the present disclosure.

[0037] FIG. 5 depicts a hardware architecture of a powder dispenser device according to an exemplary embodiment of an aspect of the present disclosure.

[0038] FIG. 6 depicts a flow chart of a learn process according to an exemplary embodiment of an aspect of the present disclosure.

[0039] FIGS. 7-10 depict flow charts of dispense processes according to exemplary embodiments of aspects of the present disclosure.

DETAILED DESCRIPTION

[0040] Aspects of the present disclosure provide a system and method for accurately and efficiently dispensing powder for loading ammunition. The dispenser or loader described herein provides a user-friendly, convenient and efficient system and method for dispensing a highly-accurate and dependable amount of powder, or charge, to fill a cartridge case. Aspects of an exemplary device are depicted in FIGS. 1-10 and described herein. The device provides a number of fixed speeds (RPM) of barrel rotation and the dispense rate (weight/time) of powder is determined for selected rotational speeds based on powder type. The terms “flow rate”, “dispense rate” and “inertial flow rate” are used interchangeably herein to refer to the amount of powder dispensed over time. As used herein the term “weight” and the like is used in its colloquial sense to refer to the amount of mass of powder. Mass is the amount of matter in a material, while weight is a measure of how the force of gravity acts upon that mass. That is, the values for mass and weight are the same when measuring on the surface of the earth.

[0041] In exemplary embodiments, the loader is configured to determine which category the powder is assigned based on a first sampling test. The type of powder is used to determine switch points, stopping points and trickle rates to approach a target weight without overshooting. Additional tests are used to determine the most efficient “high” rotational speed to deliver an amount of the powder to approach a target weight. The device is further configured to determine a speed profile that selects among high, medium and slow rotational speeds and their durations to approach the

stopping point where the dispensing switches from rotational delivery to trickle delivery for precise dispensing of powder.

[0042] In exemplary embodiments, the dispenser has twenty-five fixed speeds. The dispenser may use ten speeds to determine the most efficient speed for dispensing at the high speed. For example, five speeds are used for a range-finding test and five additional speeds centered around the best speed in the range-finding test to determine the most efficient speed.

[0043] FIGS. 1 and 2 depict a powder dispenser device 100 according to one aspect of the present disclosure. The device 100 may include a housing 102, a hopper 104 for powder storage, a cover 105, a user interface 116, and a barrel arrangement 113 for dispensing a weight of powder to a pan or dish (not shown) contemporaneously weighed on a high-accuracy scale plate 110. A digital user-interface 116 may be located on the first portion 103 of the housing 102, and used to program the device 100, initiate a dispensing operation, and display the current scale measurements and settings to the user.

[0044] With continued reference to FIGS. 1 and 2, the housing 102 may further include a second portion 108, the surface of which may include a scale plate 110 included in, and operatively coupled to a high-accuracy scale mechanism disposed within the housing 102. The second portion 108 may further include a leveling device 111. The leveling device 111 may include a volume partially filled with a liquid and an air bubble. The leveling device 111 may be configured such that when the housing 102 is level on a surface, the air bubble comes to rest at the center of a window of the leveling device, as viewed by the user. Leveling legs (not shown) may be coupled to the housing via a threaded screwing arrangement, or the like, that allow the leveling legs to move up and down independently to place the housing at a level resting angle should the surface on which the device 100 is resting not be level. The leveling legs may be used to adjust the height of one or more portions of the housing 102 to place the scale in a level space, thus increasing the accuracy of the scale. While a bubble level is described herein, one of skill in the art will recognize that other leveling functions may be implemented, including for example a laser level, without deviating from the scope of the disclosure. The second portion 108 internally may include scale circuitry and mechanics in connection with the scale plate 110 to provide a highly accurate weight of a volume of dispensed powder, as described herein. The scale circuitry and mechanics are further electronically coupled to the user interface 116 to receive and transmit operational settings and information, described herein. A transparent hood 109 is configured to be removably engaged to the second portion 108 to prevent ambient air currents from influencing the dispensing of powder and allow a user to remove the dispenser from the scale plate. In embodiments, the hood 109 may rest on the top of second portion 108 and over the scale plate 110. In alternative embodiments, the hood may be hingedly engaged to the housing 102. The hood 109 may be a transparent material, such as a plastic, or other polymer-based substance that allows a user to view the scale plate 110 under the hood 109.

[0045] A third portion 106 of the housing 102 may include the hopper 104 and the cover 105 sized and shaped to retain a volume of powder when inserted into a reservoir 107 defined by the housing 102. The hopper 104 may be a

transparent material, such as a plastic, or other polymer-based substance that allows a user to view the current volume of powder resident in the hopper 104 and reservoir 107. The hopper 104 may be sized and shaped to be just smaller than an internal circumference or perimeter of the reservoir such that the hopper 104 rests inside the reservoir 107 via a friction fit, for easy removal and replacement. Alternatively, the hopper may be affixed to the housing in any number of known methods, including, screws, clamps, or the like. The cover 105 may be sized and shaped to provide a sealing, friction fit with the hopper 104 in order to minimize the amount of air flow into and out of the hopper. The cover may also be coupled to the hopper by other mechanisms known in the art.

[0046] The third portion 106 of the housing 102 further includes a barrel arrangement 113 configured to dispense powder from the reservoir 107 and hopper 104 to a receptacle or tray disposed on the scale plate 110. A single barrel is illustrated herein, however, it is understood that a dual-barrel arrangement may be included in some embodiments. The barrel arrangement 113 may include a first barrel 112, which is in fluid communication with the reservoir 107 and extend out of the housing 102 above the second portion 108 and above the scale plate 110. The barrel 112 may include a threaded interior surface (not shown). In some embodiments, a second barrel is present, which is in fluid communication with the reservoir 107 and extends out of the housing 102 above the second portion 108 and above the scale plate 110. Preferably, a single (first) barrel 112 and a second barrel, if present, are both oriented horizontally. When a single barrel is used, it can be rotated at selected revolutions per minute (rpm) to dispense powder at high, medium, slow and trickle speeds as discussed further below. A single motor may be operatively coupled to the single barrel 110. Alternatively, an additional motor may turn the barrel at a trickle rate. A second barrel, when present, may be used to dispense powder at a trickle rate after the dispensed powder approaches a target weight. The second barrel may be operatively coupled to the same motor that turns the first barrel, by employing a second transmission. In other embodiments, a second motor turns the second barrel at a trickle rate. Generally herein, the methods described below refer to the device illustrated in FIGS. 1, 2 and 3, wherein a single barrel is present, which is preferably turned by a single motor. However, it is to be understood that the method can be practiced using any combination of motors and barrels as described above.

[0047] The third portion 106 of the housing 102 may further include a drain assembly 132 to provide a drain path for emptying of the hopper 104 and reservoir 107. The barrel arrangement and/or drain assembly may be as described in U.S. Pat. No. 10,982,938, issued Apr. 20, 2021, which is incorporated by reference it is entirety herein for all purposes.

[0048] The electronics and mechanics of the device 100 may be coupled to a power source by a power supply line (not shown). The power supply line may be configured to receive power from a source including an alternating current (AC) source, a direct current (DC) source, or the like. When the power supply line is connected to a power source, the operation of the device 100 may be initiated by activating the power button 118.

[0049] FIG. 3 depicts an expanded view of the user-interface 116 according to an exemplary embodiment of an

aspect of the disclosure. FIG. 3 illustrates the user interface 116, that can include a power button 118, a “Go” button 120, a “Learn” button 121 and a display 122. The power button 118 may activate the internal electronics and mechanics described herein via a power supply, internal or external. The “Go” button 120 can be configured to initiate a dispensing operation according to the parameters and settings input by a user or stored in a memory. The “Learn” button 121 can be configured to initiate a learning process according to the parameters and settings input by a user or stored in a memory, to be described in greater detail herein. The display 122 may include a keypad 124, one or more function buttons 126, as well as a visual output indicating the details associated with a dispensing operation. The display 122 may be an LCD, LED, OLED, or the like. The keypad 124, the power button 118, the “Go” button 120, the “Learn” button 121, function buttons 126 and others may include physical buttons electrically coupled to the circuitry, or may include touch sensitive indicators on the display 122. The visual output may include, without limitation, current device settings, current weight measured on the scale plate 110, or other indicators in connection with the dispensing and weighing of powder. The user interface 116 may also be coupled to the housing in a pivoting arrangement such that the user-interface may be rotated up or down to improve a user’s viewing angle.

[0050] At least some of the function buttons 126 may be used to control dispensing of powder when a user wishes to use manual control of the dispensing instead of using the autonomous dispensing methods described herein. According to an exemplary embodiment of an aspect of the present disclosure, the function buttons 126 may include a button for a zero function mode, in which the scale may be zeroed and a mode button for setting a dispensing mode or other mode of operation as described herein. The function buttons 126 may also include a powder function button configured to select one of several different types of powder, i.e., flake, ball, extruded, medium extruded, large extruded, or the like. A trickle function button may be configured to activate manually the rotation of a first barrel or an optional second barrel at a trickle rate as described further below, allowing a small amount of powder to be dispensed to reach a final target rate.

[0051] A calibration button 127 may be included to allow the user to calibrate and re-calibrate the scale and device. Entering calibration mode may include placing an object of known weight on the scale plate 110 and adjust the scale via the user interface 116 to match the known weight. The user interface 116 may further include a keypad 124 and one or more function buttons 126. The power button 118 and the “Go” button 129 may be hard-coded and raised buttons while the keypad 124, function buttons 126 the unit button 125 and the calibration button 127 may be integrated into the display 122 as touch-sensitive pads. Alternative configurations, however, may be implemented including varying combinations of hard-coded and touch-sensitive buttons.

[0052] FIG. 4 depicts a powder receptacle 150 according to one aspect of the disclosure for receiving dispensed powder and moving it from the scale plate to load a cartridge. The powder receptacle 150 may be formed from a lightweight metal, plastic or other material and include a handle portion 152 and one or more spout portions 154. The handle portion 152 may include a raised pattern or grip to facilitate handling of the receptacle and the powder dis-

pensed into it. The spout portions **154** may be size and shaped to facilitate the flow of powder out of the receptacle **150** and into a casing with minimal or no spillage. The receptacle **150** as described herein is exemplary and the shape and material of the receptacle may vary without deviating from the scope of the disclosure.

[0053] In operation, the device **100** may dispense a highly-accurate weight of powder through the barrel arrangement **113**. The hopper **104** and reservoir **107** may be filled with the desired powder and the cover **105** placed to seal the reservoir **107** from outside contamination. With the reservoir filled, the user may power-on the device using the power button **118** on the user interface **116**. A user may opt to zero the scale before beginning a dispensing operation. Zeroing the scale may include placing the receptacle on the scale plate **110** and pressing a “Zero” button, among the function buttons **126**. The scale will be set to a zero weight, accounting for the weight of the receptacle, such that the weight registered by the scale represents the weight of the powder alone, and not the powder and receptacle together.

[0054] Further using the user interface **116**, the user may input the desired mode, and other settings, as described herein, and press the “Go” button **120**. The device, based on the input parameters will begin to dispense the powder from the reservoir **107** through the barrel arrangement **113**. One or more motors may be disposed within the housing and coupled to the barrel **112**. The motors may drive rotation of the barrels such that powder from the reservoir **107** may flow into the internal portion **121** of the barrel **112**. The rotation of the barrel may create a flow of powder from the reservoir, through the barrels and into the receptacle resting on the scale plate. The speed at which the barrels rotate may dictate the rate at which the powder flows from the reservoir into the barrels and out to the receptacle.

[0055] The device **100** may include a computing system **1000**, as depicted in FIG. 5, to receive inputs and drive the operation of the device **100**. In general, the computing system **1000** may include a computing device **1010**, such as a special-purpose computer designed and implemented for directing and controlling the operation and provision of powder. The computing device **1010** may be or include data sources, client devices, and so forth. For example, the computing device may include the components depicted schematically in FIG. 5.

[0056] **1010** may include a microprocessor installed and disposed within a device. In certain aspects, the computing device **1010** may be implemented using hardware or a combination of software and hardware.

[0057] The computing device **1010** may communicate across a network **1002**. The network **1002** may include any data network(s) or internetwork(s) suitable for communicating data and control information among participants in the computing system **1000**. This may include public networks such as the Internet, private networks, and telecommunications networks such as the Public Switched Telephone Network or cellular networks using cellular technology and/or other technologies, as well as any of a variety other local area networks or enterprise networks, along with any switches, routers, hubs, gateways, and the like that might be used to carry data among participants in the computing system **1000**. The network **1002** may also include a combination of data networks and need not be limited to a strictly public or private network.

[0058] The computing device **1010** may communicate with an external device **1004**. The external device **1004** may be any computer or other remote resource that connects to the computing device **1010** through the network **1002**.

[0059] In general, the computing device **1010** may include a processor **1012**, a memory **1014**, a network interface **1016**, a data store **1018**, and one or more input/output interfaces **1020**. The computing device **1010** may further include or be in communication with peripherals **1022** and other external input/output devices that might connect to the input/output interfaces **1020**.

[0060] The processor **1012** may be any processor or other processing circuitry capable of processing instructions for execution within the computing device **1010** or computing system **1000**. The processor **1012** may include a single-threaded processor, a multi-threaded processor, a multi-core processor and so forth. The processor **1012** may be capable of processing instructions stored in the memory **1014** or the data store **1018**.

[0061] The memory **1014** may store information within the computing device **1010**. The memory **1014** may include any volatile or non-volatile memory or other computer-readable medium, including without limitation a Random-Access Memory (RAM), a flash memory, a Read Only Memory (ROM), a Programmable Read-only Memory (PROM), an Erasable PROM (EPROM), registers, and so forth. The memory **1014** may store program instructions, program data, executables, and other software and data useful for controlling operation of the computing device **1010** and configuring the computing device **1010** to perform functions for a user. The memory **1014** may include a number of different stages and types of memory for different aspects of operation of the computing device **1010**. For example, a processor may include on-board memory and/or cache for faster access to certain data or instructions, and a separate, main memory or the like may be included to expand memory capacity as desired. All such memory types may be a part of the memory **1014** as contemplated herein.

[0062] The memory **1014** may, in general, include a non-volatile computer readable medium containing computer code that, when executed by the computing device **1010** creates an execution environment for a computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of the foregoing, and that performs some or all of the steps set forth in the various flow charts and other algorithmic descriptions set forth herein. While a single memory **1014** is depicted, it will be understood that any number of memories may be usefully incorporated into the computing device **1010**.

[0063] The network interface **1016** may include any hardware and/or software for connecting the computing device **1010** in a communicating relationship with other resources through the network **1002**. This may include remote resources accessible through the Internet, as well as local resources available using short range communications protocols using, e.g., physical connections (e.g. Ethernet), radio frequency communications (e.g., Wi-Fi, Bluetooth), optical communications, (e.g., fiber optics, infrared, or the like), ultrasonic communications, or any combination of these or other media that might be used to carry data between the computing device **1010** and other devices. The network interface **1016** may, for example, include a router, a modem, a network card, an infrared transceiver, a radio frequency

(RF) transceiver, a near field communications interface, a radio-frequency identification (RFID) tag reader, or any other data reading or writing resource or the like.

[0064] The network interface **1016** may include any combination of hardware and software suitable for coupling the components of the computing device **1010** to other computing or communications resources. By way of example and not limitation, this may include electronics for a wired or wireless Ethernet connection operating according to the IEEE **802.11** standard (or any variation thereof), or any other short or long range wireless networking components or the like. This may include hardware for short range data communications such as Bluetooth or an infrared transceiver, which may be used to couple to other local devices, or to connect to a local area network or the like that is in turn coupled to a data network **1002** such as the Internet. This may also include hardware/software for a WiMax connection or a cellular network connection (using, e.g., CDMA, GSM, LTE, or any other suitable protocol or combination of protocols). The network interface **1016** may be included as part of the input/output devices **820** or vice-versa.

[0065] The data store **1018** may be any internal memory store providing a computer-readable medium such as a disk drive, an optical drive, a magnetic drive, a flash drive, or other device capable of providing mass storage for the computing device **1010**. The data store **1018** may store computer readable instructions, data structures, program modules, and other data for the computing device **1010** or computing system **1000** in a non-volatile form for relatively long-term, persistent storage and subsequent retrieval and use. For example, the data store **818** may store an operating system, application programs, program data, databases, files, and other program modules or other software objects and the like.

[0066] The input/output interface **1020** may support input from and output to other devices that couples to the computing device **1010**. This may, for example, include the user interface, serial ports (e.g., RS-232 ports), universal serial bus (USB) ports, optical ports, Ethernet ports, telephone ports, audio jacks, component audio/video inputs, HDMI ports, and so forth, any of which might be used to form wired connections to other local devices. This may also include an infrared interface, RF interface, magnetic card reader, or other input/output system for wirelessly coupling in a communicating relationship with other local devices. It will be understood that, while the network interface **1016** for network communications is described separately from the input/output interface **1020** for local device communications, these two interfaces may be the same, or may share functionality, such as where a USB port is used to attach to a Wi-Fi accessory, or where an Ethernet connection is used to couple to a local network attached storage.

[0067] The computing device may include or communicate with motors **1028** to drive the physical operation of the barrel assembly. The motors **1028** may be driven by the processor **1012** according to parameters and inputs from the user, user interface and input/output **1020**. The motors may also operate in conjunction with the scale **1024**, which may be operatively connected to the processor to measure and record weight measurements from the scale plate **110** during a dispensing operation. The scale **1024** may include an electromagnetic type or load cell type scale. The processor

1012 may drive the motors **1028** to dispense powder responsive to readings from the scale **1024** to reach a desired target weight of powder.

[0068] A peripheral **1022** may include any device used to provide information to or receive information from the computing device **1010**. This may include human input/output (I/O) devices such as a keyboard, a mouse, a mouse pad, a track ball, a joystick, a microphone, a foot pedal, a camera, a touch screen, a scanner, mobile device or other device that might be employed by the user **1030** to provide input to the computing device **1010** via the user interface. This may also or instead include a display, a printer, a projector, a headset or any other audiovisual device for presenting information to a user. The peripheral **1022** may also or instead include a digital signal processing device, an actuator, or other device to support control of or communication with other devices or components. In one aspect, the peripheral **1022** may serve as the network interface **1016**, such as with a USB device configured to provide communications via short range (e.g., Bluetooth, Wi-Fi, Infrared, RF, or the like) or long range (e.g., cellular data or WiMax) communications protocols. In another aspect, the peripheral **1022** may augment operation of the computing device **1010** with additional functions or features, such as a global positioning system (GPS) device, or other device. In another aspect, the peripheral **1022** may include a storage device such as a flash card, USB drive, or other solid-state device, or an optical drive, a magnetic drive, a disk drive, or other device or combination of devices suitable for bulk storage. More generally, any device or combination of devices suitable for use with the computing system **1000** may be used as a peripheral **1022** as contemplated herein.

[0069] Other hardware **1026** may be incorporated into the computing system **1000** such as a coprocessor, a digital signal processing system, a math co-processor, a graphics engine, a video driver, a camera, a microphone, speakers, and so forth. The other hardware **1026** may also or instead include expanded input/output ports, extra memory, additional drives, and so forth.

[0070] A bus **1032** or combination of busses may serve as an electromechanical backbone for interconnecting components of the computing device **1010** such as the processor **1012**, memory **1014**, network interface **1016**, other hardware **1026**, data store **1018** input/output interface **1020**, and motors **1028**. As shown in the figure, each of the components of the computing device **1010** may be interconnected using a system bus **1032** in a communicating relationship for sharing controls, commands, data, power, and so forth. In non-limiting embodiments, the memory may have settings for up to 50 dispensing protocols based on powder type and target weights.

[0071] A Learn process is described herein with reference to FIG. **6**. In an early stage in the process, powder is loaded into the hopper **104**, and the cover **105** secured to the hopper. The Learn process is initiated by depressing the “Learn” button **121** on the user interface. In some embodiments, the Learn process **200** can include three stages, or “tests.”

[0072] At Test 1 (Step 202), the motor may be run at two known speeds to dispense the powder and predict the type of gunpowder used, designated “X.” For example, Test 1 includes priming the dispensing tube, e.g., rotating the barrel at 312 RPM for 10 seconds, then rotating the barrel at 140 RPM for 10 seconds. The powder is dispensed at the known speed (e.g. 140 RPM) for 10 seconds. The time taken to

deliver the unspecified weight at the known rotational speed is indicative of the type of powder loaded in the dispenser. In embodiments, depending on the dispense rate determined during Test 1, the processor will classify powders to one of two types: Quick type (Ball, Extruded or Large Extruded) or Slow type (Flake). The weights of different types of powder for a given speed and duration may be stored in a look-up table or other memory to compare results from Test 1. At the end of Test 1, the user is prompted at step 204 to empty the powder receptacle 150. The word “EMPTY” is displayed on the display 122. The system then waits for the user to depress the “ENTER” key 128 on Display 126.

[0073] Test 1 determines whether the powder is a slow or quick dispense powder, which is then used to set the trickle “rate” and safe stop point, designated “Z”, (which are linked) based on either the slow or quick type of powder. After Test 1, the Test 1 result value will show on the display, which will classify powders to 2 types: Quick type (Ball, Extruded or Lg. Extruded) or Slow type (Flake). There are two separate sets of settings for trickles and safe stop point (Z), which are fixed regardless of target weight. That is, the slow trickle (Ts) and stop point (Zs) are selected for slow powders or quick trickle (Tq) and stop point (Zq) are selected for quick powders based on the results of Test 1. The safe stop point is a defined weight, based on the type of powder, that will be subtracted from a target weight to define the point where rotation of the barrel(s) are switched from rotational dispensing speeds to a trickle regime. Slow type (Flake) generally does not overcharge so the slow trickle regime provides large trickle angles and small safe stop point. Quick type (Ball, Extruded or Lg. Extruded) would be trickled with smaller trickle angles and larger safe stop points to minimize the chance of overcharging.

[0074] The Learn process 200 continues with Test 2 (Step 206), in which a predetermined high speed and a predetermined slow speed are used to find the speed range, designated “LRN_M” through “LRN_L” based on the inertial flow rate or dispense rate of the powder, the weight of powder delivered for a time at a given rotation rate (speed).

[0075] In some embodiments, the tube or barrel 112 will rotate at high RPM’s, e.g., for 5 seconds and pause for 5 seconds, followed by a second session slightly slower for 5 seconds and pause for 5 seconds, a third session at medium RPM for 5 seconds and pause for 5 seconds, a fourth session at a slower RPM for 5 seconds and pause for 5 seconds, and a fifth session at a slow RPM for 5 seconds. Five fixed speeds (for example, 350, 300, 250, 200 and 150 RPM) running from high to low may be used for Test 2. In another embodiment, five successively increasing RPMs may be used, e.g., 274 RPM, 324 RPM, 365 RPM, 405 RPM and 430 RPM. There may be no practical difference whether the speeds are tested from high to low or low to high, or even randomly among the five speeds selected. High to low speed ranging in Test 2 is generally used to ensure powder has filled up the tube. Without being bound by theory, it has been found that dispense rate increases as RPM increases until a maximum efficient dispense rate is reached, where the dispense rate levels off or may even decline. At the end of Test 2, the user is prompted at step 208 to empty the powder receptacle 150. The word “EMPTY” is displayed on the display 122. The system then waits for the user to depress the “ENTER” button.

[0076] At Test 3 (Step 210) the learn process determines the maximum efficient speed within the high speed range,

designated “LRN_H.” In Test 3, the processor will select a narrower range of speeds centered around the best dispense rate determined in Test 2. For example, if Test 2 determines a speed of 250 RPM has the highest effective dispense rate, Test 3 may use 230, 240, 250, 260, 270 RPM running from low to high. In some embodiments, the tube or barrel 112 will rotate at a first RPM for 5 seconds, followed by a second session slightly faster for 5 seconds, a third session slightly faster for 5 seconds, a fourth session slightly faster for 5 seconds, and a fifth session slightly faster for 5 seconds. In Test 2 and Test 3, the processor finds the best (most efficient) high speed for dispensing the powder. Medium speed (rpm=140) and slow speed (rpm 40) are fixed. That means that if the target weight is small, it will not run the best high speed, and it will select medium speed, slow speed or trickles depending on how far it is away from the target weight, as discussed below. At the end of Test 3, the user is prompted at step 212 to empty the empty powder receptacle 150. The word “EMPTY” is displayed on the display 122. The system then waits for the user to depress the “ENTER” button 128. The Learn process is completed.

[0077] If LEARN is run, the barrel would not need to be primed because the Tests 1, 2 and 3 have already filled the barrel. If LEARN is not run and the barrel is empty, the first dispensed charge will take longer because the barrel is empty. All succeeding dispenses will be more consistent.

[0078] After running the LEARN process, the results are saved in local LEARN. User 1030 can edit or add to MEM or select LEARN as mentioned in the instructions. In embodiments, “DISP” will flash and then the display will show “Normal”. Using the MODE key, one can toggle between “Normal” and “Learn” and press ENTER to save and end-configure New Memory. The dispenser will display the number of the memory that has been saved (e.g. NO 001-50). The User can press “GO” to start dispensing the saved memory or press ENTER and the dispenser will flash through all the parameters of the saved memory and stop at NO 0001-50.

[0079] If LEARN is selected, the parameters of local LEARN will be saved in, for example, MEM002 separately and independent of local LEARN. So when the user runs the next LEARN process, new local LEARN parameters will be created, but that will not affect MEM002. So a user can recall MEM002 whenever he needs. Changing environmental factors (e.g. temperature, humidity) may mean a user may choose to do LEARN at each session unless he has a generally stable environment for his loading, in which case he may recall a saved procedure from memory. A user may choose to re-do LEARN again whenever it does not run well and then save those results into memory such as MEM 001 to MEM 050.

[0080] FIGS. 7-10 illustrate the Dispense process 300 used to dispense the powder after the Learn process 200 has been completed. At step 302, the dispenser 100 is placed in “weighing mode.” At step 304, the user interface 116 waits for T, the target weight to be entered. Users choose which dispensing mode in the MODE menu setting before they start to dispense. When users press [LEARN] key for 3 sec to run LEARN process, the loader starts Test 1, Test 2, Test 3 and then shows “END” at the end of LEARN process. It is automatically set to LEARN dispensing mode. So after BOX 304 “enter target weight” and BOX 306 “press [Go] key”, the whole dispensing process is handled by the loader until/unless users press [CANCEL] to interrupt. In Block

336, loader automatically selects LEARN or NORMAL route depending on the loader's dispensing mode setting. The user depresses the "GO" button **120** at step **306** to begin the dispensing. The initial sampling process **308** is initiated. In example embodiments, while in dispensing mode, the processor will take a quick sample in the first three seconds. The sampling allows the processor to calculate the theoretical switch points for medium speed, slow speed, stop point for slow speed (or start point for long trickle) and start point for short trickle as described in the F1 process. Those calculations are based on parameters of quick or slow powder type. If the target weight T is greater than a pre-determined threshold for high speed sampling W1 (Step **310**), a predetermined high speed RPM_H is used for sampling (Step **312**). If the target weight T is less than or equal to the pre-determined threshold for high speed sampling W1 and less than the pre-determined threshold for medium speed sampling W2 (Step **314**), a predetermined medium speed RPM_M is used for sampling (Step **316**). If the target weight T is less than or equal to the pre-determined threshold for medium speed sampling W2 and less than the pre-determined threshold for slow sampling W3 (Step **318**), a predetermined slow speed RPM_L is used for sampling (Step **320**). If the target weight T is less than or equal to the pre-determined threshold for slow sampling W3 (Step **322**), no sampling is performed and default switch points are used. In embodiments, W1, W2 and W3 are arbitrary and unchanging. They are independent of quick or slow powder type from Test 1. In LEARN, the three speeds are high (best RPM determined from Test 2 and Test 3), medium (140 rpm) and slow (40 rpm). If $T > W1$, that means the target weight is very large so it is safe to sample at the best RPM determined in Test 3. The processor calculates the current dispense rate at best rpm. If $T \leq W3$, that means target weight is very small and does not have time to sample (or calculate the current dispense rate) and the processor will use the default switch points.

[0081] If $W2 > T > W3$, that means target weight is small but still has a chance to sample in slow speed to calculate the current dispense rate at slow speed. In embodiments, slow speed is always 40 rpm. The processor calculates the current dispense rate at slow speed. From the dispense rate here, it determines switch points for slow speed, long trickle and short trickle. It will run slow speed until reaching the slow speed stop point (or the start point long trickle).

[0082] Based on the relation between the target weight T and the weight thresholds in boxes **310**, **314**, **318** and **322**, a sampling at a selected speed is done in one of boxes **312**, **316**, **320** and **324**. In embodiments, the sampling time is fixed at three seconds. Once the target weight is reached, dispensing is completed for that run.

[0083] Box **326** determines switch points mainly based on the dispense rate calculated during sampling and the current gap to target weight (T-W). Sampling continues and the dispense rate is calculated even in medium speed and slow; therefore switch points are continuously updated.

[0084] For example, if target weight T is > 8 grains (gn), the dispenser will run at the high speed to check its dispense rate or double check on quick/slow powder type. Then it will run at the maximum speed determined from Test 3 and switch to medium speed and then slow speed based on the weighing parameters. If target weight T is ≤ 8 gn, it will sample at medium speed.

[0085] At step **326**, the switch points are determined, i.e., SP_M switch point to medium speed, SP_L switch point to slow speed and SP_trickle, switch point to trickles.

[0086] With reference to FIG. 8, the process includes a check/interrupt process F2 (Step **328**). The user interface **116** checks at step **330** to determine if the "CANCEL" key has been depressed. If it has been depressed, the dispensing process is interrupted at step **332**. This allows users to interrupt the dispensing process by pressing the [CANCEL] key. Users may initiate interrupt/cancel to switch to another powder or to change to a new target weight while dispensing in AUTO mode (as shown in box **332**). When dispensing is interrupted by pressing [CANCEL], it returns to weighing mode. If users want to keep dispensing with previous target weight, users can press [Go] key to resume to dispensing. Or users can enter a new target weight and start dispensing with the new target weight. If the "CANCEL" key has not been depressed, the weight is calculated at step **334**. Weight is continuously calculated and updated during dispensing. Before step **336**, NORMAL and LEARN follows the same steps. At step **336**, the processor checks to determine whether the user has selected the dispensing mode. If the user has selected the "LEARN DISPENSING MODE F3," the process continues to step **338** and moves to "B" in FIG. 9. If the user had not selected the "LEARN DISPENSING MODE F3," the process continues to step **340** and moves to "C" for the NORMAL dispensing process F4 shown in FIG. 10.

[0087] The LEARN DISPENSING MODE F3 is illustrated in FIG. 9. After the motor is stopped, some powder may keep rolling down the tube or barrel, so Z is an early safe stop point to account for any powder dispensed after the motor is stopped to minimize overweight dispenses. The stop point stops rotation of the barrel(s) and then starts turning the barrel(s) using a trickle rate. In embodiments with a single barrel, this means that the at least one motor stops rotating the barrel and then, after a brief pause to allow powder to stop dispersing from the barrel, restarts turning the barrel at a trickle rate. As discussed above, a single motor may rotate the barrel at the high, medium, and slow speeds and during the trickle rates. In other embodiments, a first motor may rotate the barrel at the high, medium, and slow speeds and a second motor may turn the barrel during the trickle rates. In dual-barrel embodiments, a first motor may rotate the first barrel at the high, medium, and slow speeds and a second motor may turn the second (trickle) barrel during the trickle rates.

[0088] At step **402**, if the weight W is greater than or equal to the difference between the target weight T and the early stop point Z determined by the LEARN process in Test 1, this means that the dispensed weight has approximately reached the target weight. The motor is stopped to end dispensing, the weight is calculated and the weight is displayed, along with the over/under status.

[0089] At step **404**, if the difference between the target weight T and the weight is greater than the flow rate/unit time D, at step **406**, the flow rate/unit time D and the predetermined deceleration curve used to determine/modify the switch points, i.e., SP_M switch point to medium speed, SP_L switch point to slow speed and SP_trickle, switch point to trickles. The deceleration curve is composed of switch points for medium speed, slow speed, long trickle, short trickle and target weight. Notably, there are no pauses at switch points, providing an uninterrupted speed profile. It

pauses only in trickles and pause(s) might get longer if scale is unstable. That means it will only trickle again only if scale is stable.

[0090] At step **408**, if the weight W is greater than or equal to $SP_trickle$, switch point to trickles, then at step **410**, the long/short trickle time is adjusted based on the powder type X (determined in Test 1). Trickles are fixed and depending on if it is Slow (flake) type or Quick (Ball, Extruded or Lg. Extruded) type. Quick and Slow are two separate but fixed sets of settings for trickles and safe stop point (Z).

[0091] At step **412**, if the weight W is greater than or equal to SP_L switch point to slow speed, then at step **414**, the slow speed LRN_L from the Learn process is used. As described above, in embodiments, the slow speed is set at 40 RPM.

[0092] At step **416**, if the weight W is greater than or equal to SP_M switch point to medium speed, then at step **418**, the medium speed LRN_M from the Learn process is used. As described above, in embodiments, the medium speed is set at 140 RPM.

[0093] Otherwise, at step **420**, the best speed LRN_H from the Learn process is used. Thereafter, the process returns to the check interrupt process at step **328** to check if the CANCEL button has been pressed by the user.

[0094] Depending on the target weight and loader operating parameters, the dispensing process desirably passes through all speed regimes: Fast, Medium, Slow, Long Trickle to Short Trickle. However, sometimes, the dispensing will pass switch points or stop/start points. Some stages may be skipped. For example, it may run at slow speed but slow speed stops very close to the target weight, so the processor skips long trickle and starts short trickles.

[0095] High, medium and slow speeds are based on the barrel continuously rotating at the indicated speed. Trickle rates are different, in that trickle rates rotate the barrel slowly for brief periods of time followed by pauses to allow for powder particles to fall onto the pan and be weighed before additional particles are dispensed. Based on the powder type determined in Test 1, trickle rates may adjust rotating time, response time and/or stop point for long and short trickles. If the tube RPM is the same for all powders, Flake powder will dispense the slowest, then Ball and Extruded are the fastest. The larger the Extruded powder the faster it will dispense. This is why powder type is important, especially for the Long (Macro) and Short (Micro) Trickle patterns. In some embodiments, the duration of rotation in trickle mode may comprise partial rotations of the barrel, so in some instances, the trickle rate is defined according to the angle the barrel is rotated through (for example, 180° , 360° , 720° , 1080° , etc.). Once the dispensed weight reaches the start point for long trickle, it starts long trickle and the energy it gives is slightly lower than slow speed. The rotating time (or angle) for long trickle will be adjusted depending how far it is away from the target weight. When the dispensed weight reaches the start point for short trickle, it starts short trickle and the energy it gives is lower than long trickle. The short trickle is the last trickle operation dispensing to desired target charge. The rotating time for short trickle is fixed and response time (or wait time) is longer than long trickle. That is because it is close to target, and it needs to wait for a weight result before the processor determines to stop or continue short trickles.

[0096] The dispensing process continues through blocks **404** to **420** depending on the weight dispensed until $W \geq T - Z$

(block **402**) is reached, at which the processor on the loader determines that further dispensing will result in an overweight status. The motor is stopped, the final weight is calculated and over/under status is displayed (blocks **422**, **424** and **426**), ending the LEARN dispensing process for that run. The user empties the receptacle (e.g. into a cartridge) and returns it to the scale pan for the next dispensing run. If the user is satisfied with the results of the LEARN process for the powder and target weight, he may save the profile into memory for future loads, or push the GO button for the loader to run the process for a subsequent dispense. As discussed above, the user may change the target weight and push the GO button for the loader to run the process for the new target weight.

[0097] The NORMAL DISPENSING MODE F4 is illustrated in FIG. 10. NORMAL mode uses default speeds, switch points and trickle profiles. The three speeds in NORMAL mode are high (300 rpm), medium (140 rpm) and slow (40 rpm). Note medium and slow speeds are the same for both NORMAL and LEARN.

[0098] At step **502**, if the weight W is greater than or equal to the target weight T , the motor is stopped to end dispensing, the weight is calculated and the weight is displayed, along with the over/under status.

[0099] At step **504**, if the difference between the target weight T and the weight W is greater than the flow rate/unit time D , at step **506**, the flow rate/unit time D is used to determine the switch points, i.e., SP_M switch point to medium speed, SP_L switch point to slow speed and $SP_trickle$, switch point to trickles.

[0100] At step **508**, if the weight W is greater than or equal to $SP_trickle$, switch point to trickles, then at step **510**, a predetermined long/short trickle time is used.

[0101] At step **512**, if the weight W is greater than or equal to SP_L switch point to slow speed, then at step **514**, a predetermined slow speed RPM_L is used.

[0102] At step **516**, if the weight W is greater than or equal to SP_M switch point to medium speed, then at step **518**, a predetermined medium speed RPM_M is used. Otherwise, at step **520**, a predetermined high speed RPM_H is used. Thereafter, the process returns to the check interrupt process at step **328**.

[0103] The dispensing process continues through blocks **504** to **520** depending on the weight dispensed until $W \geq T - Z$ (block **502**) is reached, at which the loader determines that further dispensing will result in an overweight status. The motor is stopped, the final weight is calculated and over/under status is displayed (blocks **522**, **524** and **526**), ending the NORMAL dispensing process for that run. The user empties the receptacle (e.g. into a cartridge) and returns it to the scale pan for the next dispensing run. If the user is satisfied with the results of the NORMAL process for the powder and target weight, he may save the profile into memory for future loads, or push the GO button for the loader to run the process for a subsequent dispense. Alternatively, the user may switch to the LEARN mode. As discussed above, the user may change the target weight and push the GO button for the loader to run the process for the new target weight in either LEARN or NORMAL mode.

[0104] Based on the teachings, one skilled in the art should appreciate that the scope of the present disclosure is intended to cover any aspect of the present disclosure, whether implemented independently of or combined with any other aspect of the present disclosure. For example, an

apparatus may be implemented or a method may be practiced using any number of the aspects set forth. In addition, the scope of the present disclosure is intended to cover such an apparatus or method practiced using other structure, functionality, or structure and functionality in addition to, or other than the various aspects of the present disclosure set forth. It should be understood that any aspect of the present disclosure may be embodied by one or more elements of a claim.

[0105] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects.

[0106] Although particular aspects are described herein, many variations and permutations of these aspects fall within the scope of the present disclosure. Although some benefits and advantages of the preferred aspects are mentioned, the scope of the present disclosure is not intended to be limited to particular benefits, uses or objectives. Rather, aspects of the present disclosure are intended to be broadly applicable to different technologies, system configurations, networks and protocols, some of which are illustrated by way of example in the figures and in the following description of the preferred aspects. The detailed description and drawings are merely illustrative of the present disclosure rather than limiting, the scope of the present disclosure being defined by the appended claims and equivalents thereof.

[0107] As used herein, the term “determining” encompasses a wide variety of actions. For example, “determining” may include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Additionally, “determining” may include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Furthermore, “determining” may include resolving, selecting, choosing, establishing, and the like.

[0108] As used herein, a phrase referring to “at least one of a list of items” refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover: a, b, c, a-b, a-c, b-c, and a-b-c.

[0109] The various illustrative logical blocks, modules and circuits described in connection with the present disclosure may be implemented or performed with a processor specially configured to perform the functions discussed in the present disclosure. The processor may be a neural network processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array signal (FPGA) or other programmable logic device (PLD), discrete gate or transistor logic, discrete hardware components or any combination thereof designed to perform the functions described herein. Alternatively, the processing system may comprise one or more neuromorphic processors for implementing the neuron models and models of neural systems described herein. The processor may be a microprocessor, controller, microcontroller, or state machine specially configured as described herein. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or such other special configuration, as described herein.

[0110] The steps of a method or algorithm described in connection with the present disclosure may be embodied

directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in storage or machine readable medium, including random access memory (RAM), read only memory (ROM), flash memory, erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), registers, a hard disk, a removable disk, a CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired program code in the form of instructions or data structures and that can be accessed by a computer. A software module may comprise a single instruction, or many instructions, and may be distributed over several different code segments, among different programs, and across multiple storage media. A storage medium may be coupled to a processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor.

[0111] The methods disclosed herein comprise one or more steps or actions for achieving the described method. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or action is specified, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

[0112] The functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in hardware, an example hardware configuration may comprise a processing system in a device. The processing system may be implemented with a bus architecture. The bus may include any number of interconnecting buses and bridges depending on the specific application of the processing system and the overall design constraints. The bus may link together various circuits including a processor, machine-readable media, and a bus interface. The bus interface may be used to connect a network adapter, among other things, to the processing system via the bus. The network adapter may be used to implement signal processing functions. For certain aspects, a user interface (e.g., keypad, display, mouse, joystick, etc.) may also be connected to the bus. The bus may also link various other circuits such as timing sources, peripherals, voltage regulators, power management circuits, and the like, which are well known in the art, and therefore, will not be described any further.

[0113] The processor may be responsible for managing the bus and processing, including the execution of software stored on the machine-readable media. Software shall be construed to mean instructions, data, or any combination thereof, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise.

[0114] In a hardware implementation, the machine-readable media may be part of the processing system separate from the processor. However, as those skilled in the art will readily appreciate, the machine-readable media, or any portion thereof, may be external to the processing system. By way of example, the machine-readable media may include a transmission line, a carrier wave modulated by data, and/or a computer product separate from the device, all which may be accessed by the processor through the bus interface. Alternatively, or in addition, the machine-readable

media, or any portion thereof, may be integrated into the processor, such as the case may be with cache and/or specialized register files. Although the various components discussed may be described as having a specific location, such as a local component, they may also be configured in various ways, such as certain components being configured as part of a distributed computing system.

[0115] The machine-readable media may comprise a number of software modules. The software modules may include a transmission module and a receiving module. Each software module may reside in a single storage device or be distributed across multiple storage devices. By way of example, a software module may be loaded into RAM from a hard drive when a triggering event occurs. During execution of the software module, the processor may load some of the instructions into cache to increase access speed. One or more cache lines may then be loaded into a special purpose register file for execution by the processor. When referring to the functionality of a software module below, it will be understood that such functionality is implemented by the processor when executing instructions from that software module. Furthermore, it should be appreciated that aspects of the present disclosure result in improvements to the functioning of the processor, computer, machine, or other system implementing such aspects.

[0116] If implemented in software, the functions may be stored or transmitted over as one or more instructions or code on a computer-readable medium. Computer-readable media include both computer storage media and communication media including any storage medium that facilitates transfer of a computer program from one place to another.

[0117] Further, it should be appreciated that modules and/or other appropriate means for performing the methods and techniques described herein can be downloaded and/or otherwise obtained by a user terminal and/or base station as applicable. For example, such a device can be coupled to a server to facilitate the transfer of means for performing the methods described herein. Alternatively, various methods described herein can be provided via storage means, such that a user terminal and/or base station can obtain the various methods upon coupling or providing the storage means to the device. Moreover, any other suitable technique for providing the methods and techniques described herein to a device can be utilized.

[0118] It is to be understood that the claims are not limited to the precise configuration and components illustrated above. Various modifications, changes, and variations may be made in the arrangement, operation, and details of the methods and apparatus described above without departing from the scope of the claims.

[0119] It is understood that the subject matter described herein is not limited to particular embodiments described, as such may, of course, vary. It is also understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting. Where a range of values is provided, it is understood that each intervening value between the upper and lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the disclosed subject matter.

1. A powder dispenser, comprising:
a hopper configured to hold powder;

- a receptacle configured to hold powder, the receptacle being connected to a scale, at least one barrel in fluid communication with the hopper;
 - at least one motor connected to the at least one barrel, the motor being configured to rotate the at least one barrel at a plurality of speeds to dispense powder from the hopper into the receptacle; and
 - a processor in data communication with the scale and the at least one motor;
- wherein:
- a first speed of the plurality of speeds is determined by a first calibration procedure;
 - a second speed of the plurality of speeds is determined by at least a second calibration procedure.
2. The dispenser according to claim 1, wherein the plurality of speeds comprises a high speed, a medium speed, a low speed, and a trickle rate.
 3. The dispenser according to claim 2, wherein the trickle rate is determined by the first calibration procedure, the first calibration procedure comprising:
 - rotating the at least one barrel at a first calibration speed to dispense powder into the receptacle for a first duration of time;
 - rotating the at least one barrel at a second calibration speed to dispense powder into the receptacle for second duration of time;
 - determining, via the scale, a weight of powder dispensed during the second duration of time;
 - determining, via the processor, a first calibration dispense rate based on said weight and the second duration of time;
 - classifying, via the processor, the powder based on the determined dispense rate; and
 - selecting a trickle rate based on the classification of powder.
 4. The dispenser according to claim 3, wherein the first calibration procedure further comprises selecting a safe stop point based on the classification of powder.
 5. The dispenser according to claim 3, wherein the high speed is determined by at least the second calibration procedure, the second calibration procedure comprising:
 - a plurality of cycles, each cycle comprising:
 - rotating the at least one barrel at a cycle speed to dispense powder from the hopper into the receptacle for a third duration of time;
 - determining, via the scale, a weight of powder dispensed during the third duration of time; and
 - determining, via the processor, a cycle dispense rate corresponding to the cycle speed based on said weight and the second duration of time;
 wherein each cycle speed is unique and is greater than the medium speed;
 - comparing, via the processor, the cycle dispense rates and determining a maximum cycle dispense rate; and
 - selecting the high speed based on the cycle speed corresponding to the maximum cycle dispense rate.
 6. The dispenser according to claim 3, wherein the high speed is determined by the second calibration procedure and a third calibration procedure, the second calibration procedure comprising:
 - a plurality of second calibration cycles, each second calibration cycle comprising:

rotating the at least one barrel at a cycle speed of a plurality of second calibration cycle speeds to dispense powder from the hopper into the receptacle for a third duration of time;

determining, via the scale, a weight of powder dispensed during the third duration of time; and

determining, via the processor, a cycle dispense rate based on said weight and the third duration of time;

wherein each cycle speed of the plurality of second calibration cycle speeds is unique and is greater than the medium speed;

comparing, via the processor, the cycle dispense rates and determining a maximum second calibration cycle dispense rate; and

selecting an interim high speed based on the cycle speed corresponding to the maximum second calibration cycle dispense rate; and

the third calibration procedure comprising:

a plurality of third calibration cycles, each third calibration cycle comprising:

rotating the at least one barrel at a cycle speed of a plurality of third calibration cycle speeds for a fourth duration of time;

determining, via the scale, a weight of powder dispensed during the fourth duration of time; and

determining, via the processor, a cycle dispense rate corresponding to the cycle speed based on said weight and the fourth duration of time;

wherein each cycle speed of the plurality of third calibration cycle speeds is unique and is greater than the medium speed;

comparing, via the processor, the cycle dispense rates and determining a maximum third calibration cycle dispense rate; and

selecting the high speed based on the cycle speed corresponding to the maximum third calibration cycle dispense rate.

7. The dispenser according to claim 6, wherein the interim high speed determined by the second calibration procedure is the median of the plurality of third calibration cycle speeds.

8. The dispenser according to claim 3, wherein the at least one barrel is a first barrel configured to rotate at the high speed, medium speed, and low speed, and a second barrel configured to rotate at the trickle rate.

9. A method of dispensing powder, the method comprising:

providing a dispenser having a hopper configured to hold powder, a receptacle configured to hold powder, the receptacle being connected to a scale, at least one barrel in fluid communication with the hopper, at least one motor connected to the at least one barrel, the motor being configured to rotate the at least one barrel at a plurality of operational speeds to dispense powder from the hopper into the receptacle, a processor in data communication with the scale and the at least one motor, and non-transitory computer memory in data communication with the processor;

performing a first calibration procedure to determine a first operational speed;

performing a second calibration procedure to determine a second operational speed; and

rotating the at least one barrel at an operational speed of the plurality of operational speeds to dispense powder from the hopper into the receptacle.

10. A method according to claim 9, wherein the first calibration procedure comprises:

rotating the at least one barrel at a first calibration speed to dispense powder from the hopper into the receptacle for a first duration of time;

determining, via the scale, a weight of powder dispensed during the first period of time;

determining, via the processor, a first calibration dispense rate based on said weight and the first duration of time;

classifying, via the processor, the powder based on the determined dispense rate; and

selecting a first operational speed based on the classification of powder.

11. The method according to claim 10, wherein classifying the powder further comprises:

providing a tabulated list of reference dispense rates in the memory, each reference dispense rate corresponding to a classification of powder;

comparing, via the processor, the first calibration dispense rate to the tabulated list of reference dispense rates; and

selecting a classification of powder based on the comparison.

12. The method according to claim 9, wherein the second calibration procedure comprises:

generating a plurality of second calibration discharge rates by:

rotating the at least one barrel at a plurality of second calibration speeds, wherein the at least one barrel is rotated at each second calibration speed to dispense powder from the hopper into the receptacle for a second duration of time;

determining, via the scale, a weight of powder dispensed for each second calibration during each second duration of time;

determining, via the processor, a corresponding second calibration dispense rate for each second calibration speed based on the corresponding weight and each second duration of time;

determining, via the processor, a maximum second calibration dispense rate based on the plurality of second calibration discharge rates;

determining, via the processor, a maximum second calibration speed based on the corresponding maximum second calibration discharge rate.

13. The method according to claim 12, wherein the second calibration procedure further comprises:

generating a plurality of third calibration discharge rates by:

rotating the at least one barrel at a plurality of third calibration speeds, wherein the at least one barrel is rotated at each third calibration speed to dispense powder from the hopper into the receptacle for a third duration of time;

determining, via the scale, a weight of powder dispensed for each third calibration during each third duration of time;

determining, via the processor, a corresponding third calibration dispense rate for each third calibration speed based on the corresponding weight and each third duration of time;

determining, via the processor, a maximum third calibration dispense rate based on the plurality of second calibration discharge rates;

determining, via the processor, a maximum third calibration speed based on the corresponding maximum second calibration discharge rate;

wherein the maximum second calibration speed is the median of the plurality of third calibration speeds.

14. The method according to claim **13**, wherein the maximum third calibration speed is the second operational speed.

15. The method according to claim **13**, further comprising storing at least one of the first operational speed and the second operational speed in a database located in the memory.

16. The method according to claim **9**, wherein the plurality of operational speeds includes a preset low speed and a preset medium speed.

17. A method of determining at least one optimal flow rate of powder from a powder dispenser, the method comprising:

providing a powder dispenser comprising:

a hopper configured to hold powder;

a receptacle configured to receive powder, the receptacle being connected to a scale,

at least one barrel in fluid communication with the hopper, the at least one barrel being rotatable at a plurality of operational speeds to dispense powder from the hopper into the receptacle; and

a processor in data communication with the scale; and determining a first optimal flow rate by:

obtaining a plurality of primary dispense rates by:

(a) rotating the at least one barrel at a primary speed of a plurality of primary speeds to dispense powder into the receptacle for a known primary dispense time;

(b) determining, via the scale, a primary weight of powder dispensed into the receptacle during the known primary dispense time;

(c) determining, via the processor, a primary dispense rate based on the measured weight of powder dispensed and the known primary dispense time;

(d) repeating steps (a) (c), wherein the primary speed is unique each repetition;

determining, via the processor, a maximum primary dispense rate and a corresponding optimal primary speed from the plurality of primary dispense rates;

obtaining a plurality of secondary dispense rates by:

(e) rotating the at least one barrel at a secondary speed of a plurality of secondary speeds to dispense powder into the receptacle for a known secondary dispense time;

(f) determining, via the scale, a secondary weight of powder dispensed into the receptacle during the known secondary dispense time;

(g) determining, via the processor, a secondary dispense rate based on the measured secondary weight of powder dispensed and the known secondary dispense time;

(h) repeating steps (e) (g), wherein the secondary speed is unique each repetition; and

determining, via the processor, a maximum secondary dispense rate and a corresponding maximum secondary speed from the plurality of secondary dispense rates;

wherein:

the first optimal flow rate is equal to the secondary maximum dispense rate; and

the optimal primary speed is greater than a minimum of the plurality of secondary speeds and less than a maximum of the plurality of secondary speeds.

18. The method according to claim **17**, further comprising determining a second optimal flow rate by:

rotating the at least one barrel at a first speed to dispense powder into the receptacle for a duration of time;

determining, via the scale, a weight of powder dispensed during the duration of time;

determining, via the processor, a dispense rate based on said weight and the second duration of time;

classifying, via the processor, the powder based on the determined dispense rate; and

determining, via the processor, the second optimal flow rate based on the classification of powder.

19. The method according to claim **18**, wherein the second optimal flow rate is less than the first optimal flow rate.

20. The method according to claim **18**, further comprising:

providing a non-transitory computer memory accessible by the processor; and

storing at least one of the first flow rate and the second flow rate in a database located in the memory, the database being selectively accessible by a user via a user interface.

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